

Database Technologies – Assignment 7

Use appropriate joins to solve following queries. Note: To solve below queries, use “sales” database.

1. Write a query that lists each order number followed by the name of the customer who made the order.

```
SELECT c.Cname, o.Onum
FROM customers c
INNER JOIN orders o
ON c.cnum = o.cnum;
```

```
W2_99546_Pratiksha>SELECT c.Cname, o.Onum
-> FROM customers c
-> INNER JOIN orders o
-> ON c.cnum = o.cnum;

+-----+-----+
| Cname  | Onum |
+-----+-----+
| Cisneros | 3001 |
| Hoffman  | 3003 |
| Pereira  | 3002 |
| Liu      | 3005 |
| Cisneros | 3006 |
| Giovanni | 3009 |
| Grass    | 3007 |
| Clemens  | 3008 |
| Grass    | 3010 |
| Clemens  | 3011 |
+-----+-----+
10 rows in set (0.01 sec)
```

2. Write a query that gives the names of both the salesperson and the customer for each order along with the order number.

```
SELECT c.Cname, o.Onum, s.Sname
FROM customers c
INNER JOIN orders o
ON c.cnum = o.cnum
INNER JOIN salespeople s
ON o.snum = s.snum;
```

```

W2_99546_Pratiksha>SELECT c.Cname, o.Onum, s.Sname
-> FROM customers c
-> INNER JOIN orders o
-> ON c.cnum = o.cnum
-> INNER JOIN salespeople s
-> ON o.snum = s.snum;

```

Cname	Onum	Sname
Hoffman	3003	Peel
Giovanni	3009	Axelrod
Liu	3005	Serres
Grass	3010	Serres
Grass	3007	Serres
Clemens	3011	Peel
Clemens	3008	Peel
Cisneros	3006	Rifkin
Cisneros	3001	Rifkin
Pereira	3002	Motika

```

10 rows in set (0.00 sec)

```

3. Write a query that produces all customers serviced by salespeople with a commission above 12%. Display the customer's name, the salesperson's name, and the salesperson's rate of commission.

```

SELECT c.cname, s.sname, s.comm
FROM customers c
INNER JOIN salespeople s
ON c.snum = s.snum
WHERE s.comm > 0.12;

```

```

W2_99546_Pratiksha>SELECT c.cname, s.sname, s.comm
-> FROM customers c
-> INNER JOIN salespeople s
-> ON c.snum = s.snum
-> WHERE s.comm > 0.12;

```

cname	sname	comm
Liu	Serres	0.13
Grass	Serres	0.13
Cisneros	Rifkin	0.15

```

3 rows in set (0.00 sec)

```

4. Write a query that calculates the commission earned by each salesperson for orders placed by customers with a rating above 100.

```

SELECT C.CNAME, S.SNAME, S.COMM FROM customers C inner join salespeople S ON
c.SNUM=s.Snum where s.comm>0.12;

```

```
W2_99546_Pratiksha>SELECT C.CNAME, S.SNAME, S.COMM FROM customers C inner join salespeople S ON c.SNUM=s.Snum where s.comm>0.12;
+-----+-----+-----+
| CNAME | SNAME | COMM |
+-----+-----+-----+
| Liu   | Serres | 0.13 |
| Grass | Serres | 0.13 |
| Cisneros | Rifkin | 0.15 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

- Write a query that produces all pairs of salespeople who are living in the same city. Exclude combinations of salespeople with themselves as well as duplicate rows with the order reversed.

Select SA.SNAME as salespeople1, s.sname as SALESPEOPLE2, s.city from salespeople sa
INNER JOIN salespeople s on SA.city=s.city where (SA.SNAME>S.SNAME);

```
W2_99546_Pratiksha>Select SA.SNAME as salespeople1, s.sname as SALESPEOPLE2, s.city from salespeople sa INNER JOIN salespeople s on SA.city=s.city where (SA.SNAME>S.SNAME);
+-----+-----+-----+
| salespeople1 | SALESPEOPLE2 | city |
+-----+-----+-----+
| Peel        | Motika       | London |
+-----+-----+-----+
1 row in set (0.00 sec)
```

Assignment 7 Note: To solve below queries, use “spj” database

- Display the Supplier name and the Quantity sold.
SELECT S.SNAME,SUM(SP.QTY) AS TOTAL FROM S INNER JOIN SP ON S.`S#`=SP.`S#` GROUP BY S.SNAME;

```

W2_99546_Pratiksha>use spj_db;
Database changed
W2_99546_Pratiksha>SELECT S.SNAME,SUM(SP.QTY) AS TOTAL FROM S INNER JOIN SP ON S.`S#`=SP.`S#` GROUP BY S.SNAME;
+-----+-----+
| SNAME | TOTAL |
+-----+-----+
| Smith | 900   |
| Jones | 3200  |
| Blake | 700   |
| Clark | 600   |
| Adams | 3300  |
+-----+-----+
5 rows in set (0.03 sec)

```

2. Display the Part name and Quantity sold.

```

SELECT P.PNAME,SUM(SP.QTY)AS TOTAL FROM P INNER JOIN SP ON P.`P#`=SP.`P#` GROUP BY
P.PNAME;

```

```

W2_99546_Pratiksha>SELECT P.PNAME,SUM(SP.QTY)AS TOTAL FROM P INNER JOIN SP ON P.`P#`=SP.`P#` GROUP BY P.PNAME;
+-----+-----+
| PNAME | TOTAL |
+-----+-----+
| Nut   | 1000  |
| Screw | 4900  |
| Cog   | 1500  |
| Bolt  | 300   |
| Cam   | 1000  |
+-----+-----+
5 rows in set (0.02 sec)

```

3. Display the Job name and Quantity sold.

```

SELECT J.JNAME, SUM(SP.QTY) AS TOTAL
FROM J
INNER JOIN SP ON K.`J#` = SP.`J#`
GROUP BY J.JNAME;

```

```

W2_99546_Pratiksha>SELECT J.JNAME, SUM(SP.QTY) AS TOTAL
-> FROM J
-> JOIN SP ON J.`J#` = SP.`J#`
-> GROUP BY J.JNAME;
+-----+-----+
| JNAME | TOTAL |
+-----+-----+
| Sorter | 800   |
| Console | 3300  |
| Punch  | 1200  |
| Reader | 500   |
| Collator | 1100  |
| Terminal | 400   |
| Tape   | 1400  |
+-----+-----+
7 rows in set (0.00 sec)

```

4. Display the Supplier name, Part name, Job name and Quantity sold.

```

SELECT SNAME, PNAME, JNAME, QTY
FROM S
INNER JOIN SP
ON S.`S#` = SP.`S#`
INNER JOIN P
ON P.`P#` = SP.`P#`
INNER JOIN J
ON J.`J#` = SP.`J#`;

```

```

W2_99546_Pratiksha>SELECT SNAME, PNAME, JNAME, QTY
-> FROM S
-> INNER JOIN SP
->     ON S.`S#` = SP.`S#`
-> INNER JOIN P
->     ON P.`P#` = SP.`P#`
-> INNER JOIN J
->     ON J.`J#` = SP.`J#`;

```

SNAME	PNAME	JNAME	QTY
Jones	Screw	Sorter	400
Blake	Screw	Sorter	200
Smith	Nut	Sorter	200
Adams	Cog	Punch	200
Blake	Screw	Punch	500
Jones	Screw	Punch	200
Jones	Screw	Punch	100
Adams	Bolt	Punch	200
Clark	Cog	Reader	300
Jones	Screw	Reader	200
Adams	Cog	Console	500
Adams	Cam	Console	400
Adams	Screw	Console	800
Jones	Screw	Console	500
Adams	Screw	Console	200
Adams	Bolt	Console	100
Smith	Nut	Console	700
Adams	Nut	Console	100
Adams	Cam	Collator	500
Jones	Screw	Collator	600
Jones	Screw	Terminal	400
Clark	Cog	Tape	300
Adams	Cog	Tape	200
Adams	Cam	Tape	100
Jones	Screw	Tape	800

25 rows in set (0.00 sec)

5. Display the Supplier name, Supplying Parts to a Job in the same City.

```

SELECT DISTINCT SNAME
FROM SP
INNER JOIN S
    ON SP.`S#` = S.`S#`
INNER JOIN J
    ON SP.`J#` = J.`J#`
WHERE S.CITY = J.CITY;

```

```

W2_99546_Pratiksha>SELECT DISTINCT SNAME
-> FROM SP
-> INNER JOIN S
->     ON SP.`S#` = S.`S#`
-> INNER JOIN J
->     ON SP.`J#` = J.`J#`
-> WHERE S.CITY = J.CITY;

+-----+
| SNAME |
+-----+
| Jones |
| Blake |
| Clark |
+-----+
3 rows in set (0.00 sec)

```

6. Display the Part name that is 'Red' in color, and the Total Quantity sold for that part.

```

SELECT PNAME,SUM(SP.QTY)
FROM SP INNER JOIN P
ON SP.`P#`=P.`P#`
WHERE COLOR= "RED"
GROUP BY PNAME;

```

```

W2_99546_Pratiksha>SELECT PNAME,SUM(SP.QTY)
-> FROM SP INNER JOIN P
-> ON SP.`P#`=P.`P#`
-> WHERE COLOR= "RED"
-> GROUP BY PNAME;

+-----+-----+
| PNAME | SUM(SP.QTY) |
+-----+-----+
| Nut   | 1000         |
| Screw | 1300         |
| Cog   | 1500         |
+-----+-----+
3 rows in set (0.00 sec)

```

7. Display all the Quantity sold by Suppliers with the Status = 20.

```

SELECT SNAME,SUM(SP.QTY)
FROM SP INNER JOIN S
ON SP.`S#`=S.`S#`
WHERE STATUS=20
GROUP BY SNAME;

```

```

W2_99546_Pratiksha>SELECT SNAME,SUM(SP.QTY)
-> FROM SP INNER JOIN S
-> ON SP.`S#`=S.`S#`
-> WHERE STATUS=20
-> GROUP BY SNAME;
+-----+-----+
| SNAME | SUM(SP.QTY) |
+-----+-----+
| Smith |          900 |
| Clark |          600 |
+-----+-----+
2 rows in set (0.00 sec)

```

8. Display all the Parts and their Total Quantity for the parts with the Weight > 14.

```

SELECT PNAME, WEIGHT, SUM(QTY)
FROM SP INNER JOIN P
ON SP.`P#`=P.`P#`
WHERE WEIGHT>14
GROUP BY PNAME, WEIGHT;

```

```

W2_99546_Pratiksha>SELECT PNAME, WEIGHT, SUM(QTY)
-> FROM SP INNER JOIN P
-> ON SP.`P#`=P.`P#`
-> WHERE WEIGHT>14
-> GROUP BY PNAME, WEIGHT;
+-----+-----+-----+
| PNAME | WEIGHT | SUM(QTY) |
+-----+-----+-----+
| Screw |      17 |       3600 |
| Cog   |      19 |       1500 |
| Bolt  |      17 |        300 |
+-----+-----+-----+
3 rows in set (0.00 sec)

```

9. Display all the Job names and City, which has bought more than 500 Parts.

```

SELECT JNAME, CITY, SUM(QTY)
FROM SP INNER JOIN J
ON SP.`J#`=J.`J#`
GROUP BY JNAME, CITY
HAVING SUM(QTY)>500;

```

```
W2_99546_Pratiksha>SELECT JNAME, CITY, SUM(QTY)
-> FROM SP INNER JOIN J
-> ON SP.`J#`=J.`J#`
-> GROUP BY JNAME, CITY
-> HAVING SUM(QTY)>500;
```

JNAME	CITY	SUM(QTY)
Sorter	Paris	800
Console	Anthens	3300
Punch	Rome	1200
Collator	London	1100
Tape	London	1400

5 rows in set (0.00 sec)

10. Display all the Part names and Quantity sold that have a Weight less than 15.

```
SELECT PNAME,SUM(QTY)
FROM SP INNER JOIN P
ON SP.`P#`=P.`P#`
WHERE WEIGHT<15
GROUP BY PNAME;
```

```
W2_99546_Pratiksha>SELECT PNAME, SUM(QTY) AS TOTAL
-> FROM SP
-> INNER JOIN P
-> ON SP.`P#` = P.`P#`
-> WHERE WEIGHT < 15
-> GROUP BY PNAME;
```

PNAME	TOTAL
Screw	3600
Cog	1500
Bolt	300

11. Display all the Suppliers with the same Status as the supplier, 'CLARK'.

```
SELECT S1.SNAME, S1.STATUS
FROM S AS S1
JOIN S AS S2
ON S1.STATUS=S2.STATUS
WHERE S2.SNAME= 'CLARK' AND S1.SNAME>S2.SNAME;
```

```

W2_99546_Pratiksha>SELECT S1.SNAME, S1.STATUS
-> FROM S AS S1
-> JOIN S AS S2
-> ON S1.STATUS=S2.STATUS
-> WHERE S2.SNAME= 'CLARK' AND S1.SNAME>S2.SNAME
+-----+-----+
| SNAME | STATUS |
+-----+-----+
| Smith |      20 |
+-----+-----+
1 row in set (0.00 sec)

```

12. Display all the Parts which have more Weight than any "Red" parts.

```

SELECT P1.PNAME,P1.COLOR,P1.WEIGHT
FROM P AS P1
JOIN P AS P2
ON P1.COLOR=P2.COLOR=P2.COLOR
WHERE P1.WEIGHT>P2.WEIGHT AND P2.COLOR= 'RED';

```

```

W2_99546_Pratiksha>SELECT P1.PNAME,P1.COLOR,P1.WEIGHT
-> FROM P AS P1
-> JOIN P AS P2
-> ON P1.COLOR=P2.COLOR=P2.COLOR
-> WHERE P1.WEIGHT>P2.WEIGHT AND P2.COLOR= 'RED';
+-----+-----+-----+
| PNAME | COLOR | WEIGHT |
+-----+-----+-----+
| Bolt  | Green |      17 |
| Bolt  | Green |      17 |
| Screw | Blue  |      17 |
| Screw | Blue  |      17 |
+-----+-----+-----+
4 rows in set, 3 warnings (0.00 sec)

```

13. Display all the Jobs going on in the same city as the job 'TAPE'.

```

SELECT J1.JNAME,J1.CITY
FROM J AS J1
FROM J AS J2
ON J1.JNAME=J2.JNAME
WHERE J1.CITY=J2.CITY AND J1.JNAME= "TAPE";

```

```
W2_99546_Pratiksha>SELECT J1.JNAME, J1.CITY
-> FROM J AS J1
-> INNER JOIN J AS J2
-> ON J1.JNAME = J2.JNAME
-> WHERE J1.CITY = J2.CITY
-> AND J1.JNAME = "Tape";
```

```
+-----+-----+
| JNAME | CITY  |
+-----+-----+
| Tape  | London|
+-----+-----+
```

```
1 row in set (0.00 sec)
```

14. Display all the Parts with Weight less than any of the "Green" parts.

```
SELECT DISTINCT P1.PNAME, P1.WEIGHT
FROM P AS P1
JOIN P AS P2
ON P1.WEIGHT < P2.WEIGHT
WHERE P2.COLOR = 'GREEN';
```

```
W2_99546_Pratiksha>SELECT DISTINCT P1.PNAME, P1.WEIGHT
-> FROM P AS P1
-> JOIN P AS P2
-> ON P1.WEIGHT < P2.WEIGHT
-> WHERE P2.COLOR = 'GREEN';
```

```
+-----+-----+
| PNAME | WEIGHT |
+-----+-----+
| Nut   | 12     |
| Screw| 14     |
| Cam   | 12     |
+-----+-----+
```

```
3 rows in set (0.00 sec)
```

15. Display the name of the Supplier who has sold the maximum Quantity (in one sale).

```
SELECT S.SNAME, MAX (SP.QTY)
FROM SP INNER JOIN S
ON SP.`S#`=S.`S#`
GROUP BY S.SNAME
ORDER BY S.SNAME
LIMIT 1;
```

```

W2_99546_Pratiksha>SELECT S.SNAME, MAX(SP.QTY) AS MAX_QTY
-> FROM SP
-> INNER JOIN S
-> ON SP.`S#` = S.`S#`
-> GROUP BY S.SNAME
-> ORDER BY S.SNAME
-> LIMIT 1;
+-----+-----+
| SNAME | MAX_QTY |
+-----+-----+
| Adams |      800 |
+-----+-----+
1 row in set (0.00 sec)

```

16. Display the name of the Supplier who has sold the maximum overall Quantity (sum of Sales).

```

SELECT S.SNAME, SUM(SP.QTY)
FROM SP INNER JOIN S
ON SP.`S#`=S.`S#`
GROUP BY S.SNAME
ORDER BY SUM(SP.QTY)DESC
LIMIT 1;

```

```

W2_99546_Pratiksha>SELECT S.SNAME, SUM(SP.QTY)
-> FROM SP INNER JOIN S
-> ON SP.`S#`=S.`S#`
-> GROUP BY S.SNAME
-> ORDER BY SUM(SP.QTY)DESC
-> LIMIT 1;
+-----+-----+
| SNAME | SUM(SP.QTY) |
+-----+-----+
| Adams |      3300 |
+-----+-----+
1 row in set (0.00 sec)

```